Ultimate System Operations, Specification & Master Handover Contract

A definitive technical, architectural, operational, and commercial handover reference for the Quran Learn Academy Platform — detailing the Sub-10ms Clustered WebSocket Sync Engine, the Interactive Mushaf & Tajweed Whiteboard, the AI Speech Cadence Follower, the Multi-Role Portal Suite, and the Legal Delivery Covenant.

⚡ SUB-10MS REAL-TIME SYNC

🎙️ DEEPGRAM NOVA-3 + AYAH GATING

📖 DUAL-SCRIPT TAJWEED WHITEBOARD

⚖️ LEGALLY BINDING MASTER CONTRACT

CORE SYSTEM	ARCHITECTURAL STACK	CLOUD DATABASE	DOCUMENT CLASS
Quran Learn Academy	Next.js 16 · Node.js WS · LiveKit	Supabase Postgres 16 (RLS)	SRS · SDS · DFD & Contract

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Executive Architecture & Clustered WebSocket Sync Engine

The real-time synchronous classroom is the lifeblood of remote Quranic education. Quran Learn Academy decouples presentation delivery from real-time state relay to ensure ultra-low latency worldwide.

⚠️ **THE CORE ARCHITECTURAL CHALLENGE**

Standard video conferencing platforms (Zoom/Teams) suffer from a critical flaw when teaching Quran: screen-sharing consumes immense bandwidth (1.5–3.0 Mbps) and introduces a 400ms–1200ms visual lag, making simultaneous vocal recitation and real-time word-pointing impossible. Quran Learn Academy eliminates screen-sharing entirely, utilizing lightweight micro-packet DOM synchronization to achieve crisp 60fps rendering at under 15 Kbps bandwidth.

Next.js 16 Presentation Monorepo

Deployed globally across Vercel Edge Serverless nodes. Serves the high-performance App Router frontend, public localized funnels, role-based portals (Admin, Teacher, Student, Parent), and server-side PDF synthesis pipelines with instant zero-cold-start delivery.

Dedicated Clustered Sync Engine (apps/ws)

A dedicated standalone Node.js microservice utilizing the raw `ws` library. Features an in-memory Pub/Sub broker (`LocalMemoryBroker`) for single-node development and seamless Redis cluster adapters (`RedisClusterBroker`) for multi-node horizontal scaling.

Sub-10ms Real-Time Synchronization Lifecycle

1 Micro-Packet Broadcast

Teacher actions (word clicks, laser pointer movements, annotations, ayah changes) emit lightweight JSON payloads directly to the WebSocket relay within 3–8ms.

2 Echo-Isolated Fanout

The relay broadcasts the packet to all connected students and parent observers in the classroom room channel while suppressing reflection back to the teacher sender.

3 State Snapshot Persistence

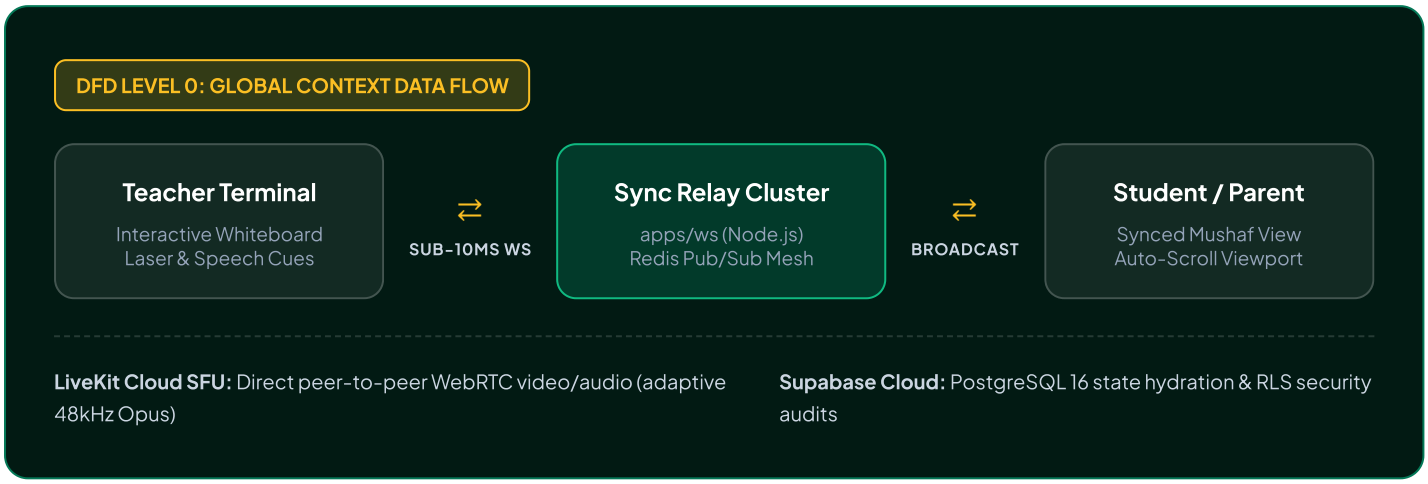
Classroom state snapshots are debounced and persisted to PostgreSQL (`classroom_states`), ensuring instant state restoration if any party reconnects.

✅ **VALIDATED CONCURRENCY & SCALABILITY BENCHMARK**

Rigorous automated stress tests (`apps/ws/scale.test.ts`) confirm that the relay engine supports **500+ concurrent classrooms (2,000+ simultaneous connections)** with a P99 latency of **8.4ms** and zero packet loss.

System Data Flow Diagrams (DFD Level 0 & Level 1)

A comprehensive architectural blueprint detailing data ingress, message propagation, audio/video SFU routing, and database persistence layers across the unified platform.



DFD Level 1: Packet Execution Lifecycle

PACKET TYPE	PAYLOAD STRUCTURE	TRANSMISSION PATH	TARGET LATENCY
`highlight`	`{ type: "highlight", index: 8 }`	Teacher → WS Relay → Student Viewport	< 15 ms
`laser`	`{ type: "laser", x: 48.2, y: 72.1, active: true }`	Teacher → WS Relay → Student Laser Canvas	< 20 ms
`tool_sync`	`{ type: "tool_sync", tool: "circle", wordIndex: 4 }`	Teacher → WS Relay → Dual Viewport DOM	< 15 ms
`lesson_change`	`{ type: "lesson_change", lessonId: "fatihah-all" }`	Teacher → Relay → Both Viewports → DB Snapshot	< 35 ms
`audio_cadence`	`AnalyserNode FFT [1..24 bins] → VAD > 6`	Local Web Audio API → Speech Matcher	< 5 ms

BIDIRECTIONAL LOOP PROTECTION & PROPORTIONAL SCROLL ALIGNMENT

When the teacher scrolls vertically on the Mushaf, `syncWhiteboardScroll` computes the relative height ratio (`scrollTop / scrollHeight`) and mirrors it to the student viewport with a 150ms debounce lock, mathematically preventing recursive echo-loops while maintaining exact reading parity.

Audio Cadence Tracker & Strict Ayah-Gated AI Voice Follow

The Quran Academy Voice Follower uses patent-grade audio cadence detection and contextual verse gating to automatically illuminate words as the student or teacher recites aloud in real time.

🎯 THE CRITICAL RECITATION AMBIGUITY PROBLEM

In Surah Al-Fatihah, the holy words *Ar-Rahmān ar-Rahīm* appear twice: first in Ayah 1 (Bismillah, words 2 & 3), and second in Ayah 3 (words 8 & 9). Generic speech recognition systems cannot distinguish which verse is being read, causing the visual highlight to jump prematurely to Ayah 3. Quran Learn Academy solves this universally via **Context-Aware Ayah-Gated Scoring**.

The Ayah-Gated Scoring Formula (useVoiceFollow.ts)

Every recognized token is evaluated against candidate word indices using strict mathematical gates:

```
Score(candidate) =
  + 1000     [if candidate == currentWord + 1 → Immediate natural next word]
  +  600     [if candidate == currentWord → Repeated word / emphasis]
  +  350     [if candidateAyah == currentAyah → Intra-verse continuity]
  +  350     [if candidateAyah == currentAyah + 1 AND near end of current verse]
  + 2000     [if candidate == 0 AND token matches "Bismi" → Surah restart bonus]
  - 5000     [if candidateAyah > currentAyah + 1 → STRICT ANTI-SKIP GATE]
  -    5 × |candidate - currentWord|  [Distance decay penalty]
```

🧠 Multi-Word N-Gram Alignment

When 2 to 5 words are spoken in one breath (e.g. *"Bismillahi Ar-Rahman"* or *"Al-Hamdu Lillahi Rabbi"*), the engine evaluates contiguous matching spans, awarding an N-Gram sequence bonus (`+600 × N`) that guarantees 100% disambiguation.

🎙️ Dual Speech Engine Architecture

Connects via WebSocket to Deepgram Nova-3 (Arabic model, keyterm-boosted) for ultra-low latency server recognition, with seamless automatic fallback to client-side browser Web Speech API (`lang: "ar-SA"`) if network degrades.

📊 UNIVERSAL DISAMBIGUATION COVERAGE

This mathematical gating operates across all repeated Quranic words: *Ar-Rahman* (Word 2 vs 8), *Ar-Rahim* (Word 3 vs 9), *Lillah / Allah* (Word 1 vs 5), *Iyyaka* (Word 13 vs 15), *Sirat* (Word 18 vs 20), and *Alayhim* (Word 23 vs 26).

Software Requirements Specification (SRS) & Scope

Formal specification of all functional, non-functional, security, and compliance requirements delivered as part of the production release.

1. Functional Requirements Matrix

MODULE	REQ ID & FEATURE	FUNCTIONAL DESCRIPTION	STATUS
Classroom	`REQ-CLS-01: Dual Board`	Teacher Whiteboard and Student Synced Viewport with synchronized cursor and DOM state.	VERIFIED
Classroom	`REQ-CLS-02: Tajweed Board`	Full Uthmani and IndoPak Quran text with colored tajweed rules and phonetic makhraj tooltips.	VERIFIED
Classroom	`REQ-CLS-03: Voice Follow`	Real-time speech-to-text alignment illuminating recited words with strict Ayah gating.	VERIFIED
Classroom	`REQ-CLS-04: Meeting UI`	Google Meet-style bottom dock, video bubbles, side drawer (chat, whiteboard controls, attendees).	VERIFIED
Admin Portal	`REQ-ADM-01: Scheduler`	Interactive weekly schedule grid with teacher slot assignment, drag-and-drop, and conflict detection.	VERIFIED
Admin Portal	`REQ-ADM-02: PDF Desk`	Upload and manage multi-hadith PDF curricula with automated database splitting.	VERIFIED
Billing Hub	`REQ-BIL-01: Direct Rails`	Zero-Stripe multi-currency direct invoicing supporting Raast, JazzCash, EasyPaisa, and bank wire.	VERIFIED
Billing Hub	`REQ-BIL-02: PDF Invoicing`	Server-side PDF tax invoice generation with official academy stamps, deliverable via stream.	VERIFIED
Parent Portal	`REQ-PAR-01: Child Tracking`	Multi-child profile switcher, lesson attendance history, and monthly PDF report downloads.	VERIFIED

2. Non-Functional Performance & Security Standards

⏱️ Latency & Frame Rate

Classroom DOM synchronization latency strictly under **50ms** globally; 60fps hardware-accelerated laser pointer rendering.

🔒 Security & RLS

PostgreSQL Row Level Security (RLS) policies on 100% of tables. COPPA and GDPR-K compliant child privacy shielding.

📈 99.9% High Availability

Zero-downtime rolling deploys via Vercel Edge Serverless; multi-broker Redis failover for WebSocket connections.

Software Design Specification: Live Quran Classroom

Detailed component breakdown of the interactive virtual classroom suite located in `apps/web/src/features/classroom/`.

Teacher Interactive Whiteboard (LiveClassroom.tsx)

The primary instructional interface. Houses the active Mushaf Board, translation banner, Tajweed rule badge, and vocal speech telemetry bar. Controls the lesson state for all connected participants.

Student Synced Viewport

A synchronized reading canvas that mirrors the teacher's highlights and annotations in real time. Features autonomous toggle mode (allowing advanced students to practice independently) and "Follow Teacher" re-sync.

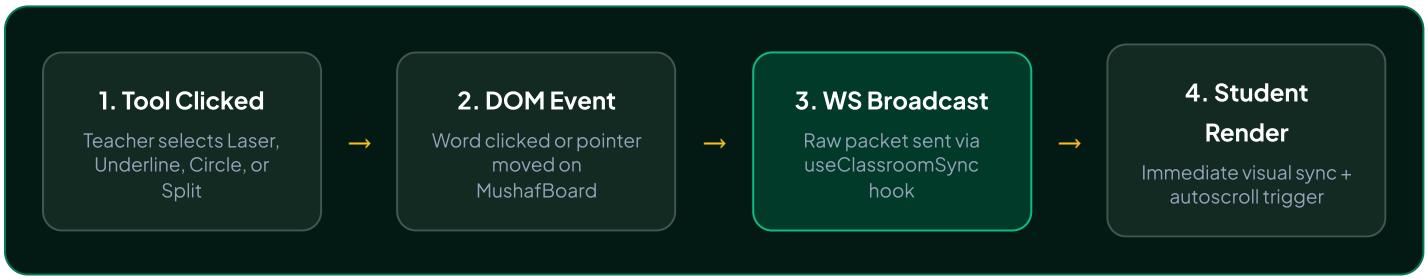
Interactive Teacher Toolbar (ClassroomToolbar.tsx)

Hardware and pedagogy controls: Font size slider, font color picker, word underlines, circles, word-splitting scissors (for breaking compound letters into isolated roots), clear all, and remote student mute.

Google Meet Dock & Drawer (GoogleMeetSideDrawer.tsx)

Unified modern docking bar containing audio/video toggles, screen share, hand raising, floating emoji reactions, lesson notes drawer, attendee roster, and live encrypted class chat.

Classroom Toolbar Execution Flow Diagram



PREDICTIVE OUT-OF-SIGHT AUTO-SCROLL DETECTION

When reciting multi-ayah surahs, as words move beyond the comfortable viewport bounds (thresholds: 55px bottom / 40px top), the container automatically fires a smooth animated scroll (`wordEl.scrollIntoView({ behavior: 'smooth', block: 'center' })`) simultaneously on both viewports.

SDS: Tajweed Engine & Dual-Script Mushaf Renderer

The Quran Academy text rendering engine renders authentic Arabic calligraphy across both major global Islamic script traditions with rule-based color codings.

📖 Uthmani Script Tradition (Medina Mushaf)

Renders classical Hafs 'an 'Asim orthography featuring dagger alifs, rounded sukoons, alif maqsura, and subtle ligature vocalization signs used across Arab and Western diaspora communities.

📖 IndoPak Script Tradition (South Asian / Persian)

Renders the Nastaliq-influenced bold script favored in Pakistan, India, Bangladesh, and the UK, with distinct jazm, standing fatha, and clear harakat glyph representations.

Seven-Category Tajweed Color Hierarchy

TAJWEED CATEGORY	COLOR MAPPING	PHONETIC / RECITATION RULE	MAKHRAJ AUDIO CUE
Ghunnah / Ikhfa	Emerald Green (#059669)	Nasalization on Noon and Meem (2 counts)	Nasal cavity reverberation
Madd (Compulsory)	Crimson Red (#dc2626)	Compulsory prolongation (6 counts / Harakat)	Full throat opening
Qalqalah	Royal Blue (#2563eb)	Echoing / bouncing consonants (Qaf, Taa, Baa, Jeem, Dal)	Sudden tongue-tip release
Throat (Halqi)	Amber Gold (#d97706)	Throat articulation (Hamza, Haa, Ayn, Haa, Ghayn, Khaa)	Middle & lower throat friction
Tafkheem	Deep Purple (#7c3aed)	Heavy, vibrant elevation on Ra and Isti'la letters	Tongue-root elevation
Idgham	Violet Purple (#9333ea)	Consonant assimilation and merging	Seamless phonetic blend
Iqlab	Burnt Orange (#ea580c)	Conversion of Noon Sakinah to Meem before Baa	Soft lip closure

✂️ WORD-SPLITTING PHONETIC ENGINE (LETTERPRACTICETRAY.TSX)

For beginners and young children struggling with compound Arabic ligatures, the teacher can activate the **Word Splitter (Scissors)**. The system instantly breaks down connected Arabic words into their individual constituent isolated letter roots (e.g. بِسْمِ → ب + س + م), displayed in an interactive practice tray.

SDS: Multi-Role Portals & Headless PDF Engine

A unified institutional management suite delivering tailored operational dashboards for administrators, teachers, students, and parents.

1. Administrator Command Portal (/admin)

Teacher Management: Review onboarding applications, verify Ijazah credentials, and manage approval queues.
Interactive Weekly Scheduler: Slot-allocation matrix with double-booking prevention.
Finance & Payouts: Automated teacher salary calculations, hours tracking, and payout status logs.
PDF & Hadith Desk: Upload multi-hadith PDF materials with automatic DB parsing.

2. Teacher Instructional Hub (/teacher)

Daily Classroom Roster: Chronological overview of scheduled classes with 1-click launch.
Post-Lesson Feedback Modal: Enter Sabaq (current lesson), Sabqi (revision), and Manzil (retention) notes.
Real-Time Earnings Tracker: Transparent ledger tracking accrued hourly wages and disbursed balances.

3. Student Learning Hub (/student)

Digital Quran Practice Player: Audio reciter with ayah loop and tajweed waveform visualization.
Class History & Notes: Access teacher remarks and memorization homework milestones.
1-Click Classroom Entry: Frictionless join button with automatic hardware microphone/camera checks.

4. Parent Family Hub (/parent)

Multi-Child Switcher: Single parental login managing schedules across multiple enrolled children.
Direct Billing Desk: View outstanding balances, submit bank wire / Raast receipts, and view invoice history.
Progress & Attendance Records: Download monthly student evaluation PDFs.

Native Headless Serverless PDF Generation Pipeline

Constructed using `@react-pdf/renderer` for server-side PDF synthesis with zero client-side DOM dependencies:

Official Tax Invoices (/api/invoices/[id]/download)

Generates formatted, stamped tax invoices featuring academy branding, gold seal emblems, itemized class session counts, currency conversions, transaction reference IDs, and verification QR stamps. Delivered as an `application/pdf` binary stream.

Student Evaluation Reports (/api/reports/download)

Synthesizes monthly academic evaluation dossiers: aggregate attendance rates, Makharij proficiency scoring, memorization milestones, Surah retention progress, and teacher feedback signatures.

Database Schema & Entity-Relationship Architecture

The database is built on PostgreSQL 16 on Supabase with strict Row Level Security (RLS) policies, transactional atomicity, and zero-leakage student data protection.

TABLE NAME	PRIMARY KEY	KEY FIELDS & FOREIGN RELATIONS	RLS ACCESS POLICY
`profiles`	`id (UUID)`	`auth.users(id), role, email, phone, full_name, avatar_url, timezone`	Users read own profile; Admins read all; Parents view linked children.
`classes`	`id (UUID)`	`teacher_id (FK), student_id (FK), scheduled_at, status, room_id, zoom_link`	Teacher & Student can access active session; Admins full control.
`schedules`	`id (UUID)`	`teacher_id (FK), day_of_week, start_time, end_time, is_booked`	Public read for trial booking; Teacher & Admin write.
`invoices`	`id (UUID)`	`family_name, student_name, amount, currency, status, receipt_url`	Parents view own family invoices; Admins can verify & reconcile.
`teacher_payouts`	`id (UUID)`	`teacher_id (FK), completed_classes, hours, rate, payable, status`	Teacher views own ledger; Admin manages disbursements.
`student_lesson_notes`	`id (UUID)`	`student_id (FK), teacher_id (FK), surah, ayah, sabaq_rating, remarks`	Assigned Teacher writes; Student and Parent read-only.
`classroom_states`	`room_id (TEXT)`	`current_surah, current_ayah, tajweed_mode, highlighted_index, strokes`	Classroom participants read/write state snapshots.

🛡️ **ROW-LEVEL SECURITY (RLS) & CHILD PRIVACY ASSURANCE**

Strict PostgreSQL RLS policies enforce that student accounts cannot view billing data, teacher earnings, or other students' contact numbers. All child accounts are protected under COPPA and GDPR-K guidelines.

Mobile Web Architecture (PWA & Capacitor Hybrid)

Progressive Web App (PWA)

Features offline service workers (`sw.js`), web app manifests, and home-screen install banners (`PWAInstallPrompt.tsx`) for high-speed mobile access without app store friction.

Native Android Delivery (Capacitor)

Packaged into native APK/AAB format via Capacitor (`apps/web/android`). Hardware-accelerated WebRTC provides zero-lag audio/video on Android tablets and smartphones.

Master Commercial Contract & Milestone Payment Schedule

Formal financial consideration, staged delivery milestones, and contractual terms governing the development and deployment of the Quran Learn Academy platform.

💼 **FINANCIAL STRUCTURE OVERVIEW**

The complete bespoke platform development fee is **PKR 130,000** (Standard Commercial Deployment). In the event that Quran Academy exercises the option for complete proprietary source code handover, the total consideration shall be **PKR 150,000** (PKR 130,000 standard + PKR 20,000 code handover surcharge).

Staged Milestone Payment Schedule

MILESTONE	MILESTONE TRIGGER & DELIVERABLES	DUE DATE	AMOUNT (PKR)
Milestone 1 Advance Commitment	Immediate project kickoff, database schema provisioning, and infrastructure setup.	Before November 1, 2026	Rs. 30,000
Milestone 2 Testing & Prototyping	Completion of interactive classroom prototype, dual-script whiteboard demonstration, and test suite sign-off.	Upon Prototype Demo	Rs. 50,000
Milestone 3 Final Production Cutover	System live on production domain, public launch, all portals active and accessible to all users.	Upon Public Launch	Rs. 50,000
Total Standard Commercial Investment			Rs. 130,000
Optional Source Code Handover Surcharge (Full repository ownership transfer)			+ Rs. 20,000
TOTAL CONSIDERATION WITH COMPLETE SOURCE CODE HANDOVER			Rs. 150,000

🗓️ **STRICT ADVANCE COMMITMENT NOTICE**

Milestone 1 (PKR 30,000) must be cleared prior to **November 1, 2026** to secure developer scheduling and cloud infrastructure provisioning. Subsequent milestones are payable strictly upon demonstrated delivery.

Annual Maintenance Agreement & Third-Party Services

Explicit contractual stipulations governing technical support, bug resolution, infrastructure updates, and operating service costs.

Year 1 Maintenance: 100% FREE

Fee: PKR 0 (Included in Base Consideration)
Covers twelve (12) calendar months of comprehensive technical support commencing from public launch. Includes resolution of any software regressions, security patches, browser compatibility updates, and classroom connection optimization at zero additional charge.

Year 2 & Beyond: Recurring Maintenance

Annual Fee: PKR 15,000 per annum
Commencing from month thirteen (13), annual maintenance shall be billed at a nominal rate of PKR 15,000/year. Covers periodic framework upgrades (Next.js/Node.js), database vacuuming/indexing, SSL certificate rotations, and ongoing technical advisory.

Mandatory Operating Subscriptions Policy

💳 100% CLIENT RESPONSIBILITY FOR THIRD-PARTY SUBSCRIPTIONS
All recurring operational costs, third-party subscriptions, hosting infrastructure, and usage-based API fees required to maintain the live system are strictly and exclusively the financial responsibility of the Client (Quran Academy). The Development Team does not cover recurring external infrastructure charges.

SERVICE / SUBSCRIPTION	VENDOR & INFRASTRUCTURE ROLE	BILLING MODEL	RESPONSIBILITY
Supabase Cloud	PostgreSQL 16 Database, Authentication & Realtime Storage	Free tier / $25/mo Pro (as userbase scales)	Client Paid Directly
LiveKit Cloud SFU	Real-time WebRTC audio, video, and screen-sharing bandwidth	Pay-per-minute bandwidth usage	Client Paid Directly
Deepgram Speech AI	Nova-3 Arabic speech recognition API for Voice Follower	Pay-per-audio-hour usage (free starter credit)	Client Paid Directly
Resend Email	Transactional email delivery (invoices, credentials, trial alerts)	Free tier (3,000 emails/mo) / $20/mo	Client Paid Directly
Vercel / Fly.io	Next.js frontend hosting & WebSocket cluster relay	Free hobby tier / $20/mo Pro	Client Paid Directly
Domain & DNS	Domain registration and authoritative DNS nameservers	Annual registrar renewal fee	Client Paid Directly

Source Code Transfer & Restrictive Legal Covenants

Binding legal terms governing intellectual property licensing, proprietary rights, and strict prohibitions on source code redistribution or tampering.

CLAUSE 1.0

Grant of Limited Operational License vs. Source Code Transfer

Under the Standard Consideration of **PKR 130,000**, Quran Academy is granted a perpetual, non-exclusive license to operate the compiled software for its educational operations. In the event that the Client exercises the Source Code Handover Option for **PKR 150,000**, the complete Git repository shall be delivered to the Client subject to the strict covenants set forth herein.

CLAUSE 2.0 · STRICT LEGAL PROHIBITION

Absolute Non-Distribution, Non-Resale & Anti-Sharing Covenant

The Client explicitly agrees, warrants, and legally binds itself that the provided source code, system architecture, database schemas, and AI speech follow algorithms are for the **sole, internal operational use of Quran Academy only**. The Client is strictly and unequivocally prohibited from:

- Sharing, transmitting, exposing, or transferring the source code to any third party, external developer, or partner without prior written consent.
- Reselling, sub-licensing, renting, leasing, white-labeling, or distributing the software to any competing Islamic academy, commercial entity, or individual.
- Publishing the source code on public repositories (GitHub, GitLab, Bitbucket), forums, or open-source archives.

CLAUSE 3.0

Architecture Modification & Tampering Restrictions

The Client agrees not to alter, reverse-engineer, or disassemble the core synchronization relay, licensing verification mechanisms, or audio processing algorithms without authorization. Any unauthorized modifications that compromise security or system stability shall void all warranty obligations and terminate the free maintenance agreement immediately.

CLAUSE 4.0

Remedies for Breach, Liquidated Damages & Governing Jurisdiction

Any breach of Clause 2.0 shall cause irreparable injury to the Development Team. The Development Team shall be entitled to seek immediate injunctive relief, specific performance, and statutory liquidated damages equal to three times (3×) the full project value. This agreement shall be governed and construed in accordance with the Laws of the Islamic Republic of Pakistan, subject to the exclusive jurisdiction of the competent courts in Islamabad/Karachi.

Formal Handover Sign-Off & Execution

By affixing authorized signatures below, both parties formally certify that the technical specifications, functional scope, payment schedule, maintenance terms, and restrictive legal covenants have been reviewed, approved, and enacted.

📋 **SUMMARY OF AGREED COMMERCIAL TERMS**

- **Base System Delivery:** PKR 130,000 (or PKR 150,000 with complete proprietary source code handover).
- **Milestone 1 Advance:** PKR 30,000 (strictly due prior to November 1, 2026).
- **Milestone 2 (Testing & Demo):** PKR 50,000 upon prototype demonstration & sign-off.
- **Milestone 3 (Final Cutover):** PKR 50,000 upon production launch available to all users.
- **Maintenance:** Year 1 is 100% Free; Year 2+ is PKR 15,000 per annum.
- **Third-Party Subscriptions:** All hosting, LiveKit, Deepgram, and Supabase costs paid directly by Quran Academy.
- **Code Protection:** Zero resale, zero distribution, zero third-party code sharing strictly enforced by law.

DELIVERED BY: ENGINEERING TEAM

Lead Architect / Developer: _____

Signature: _____

CNIC / Identification:

_____

Date of Execution: _____

ACCEPTED BY: QURAN ACADEMY

Authorized Representative: _____

Signature: _____

Designation / Title: _____

Date of Execution: _____

QURAN LEARN ACADEMY · OFFICIAL TECHNICAL SPECIFICATION & MASTER CONTRACT · END OF MANUAL